ReadWise AI — Pediatric Dyslexia Risk Screening Report

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Child Name: Arun	**Age:** 9 years	**Grade:** Grade 4
Assessment Task: T4_Meaningful_Text	**Primary Language:** English	**Session ID:** #1009

AI SCREENING ESTIMATE: HIGH DYSLEXIA RISK INDICATOR

Estimated Risk Probability: **78.0%** | Classification Confidence: **89.0%**

IMPORTANT NOTICE: *This screening result is an educational early-warning assessment designed to triage support. It does NOT constitute a formal clinical or medical diagnosis of dyslexia.*

1. Multimodal Behavioral Analytics Summary

Modality	Observed Metric	Measured Value	Typical Cohort Reference	Status
Eye-Tracking	Gaze Regressions	32 events	< 18 events	Elevated Backtracking
Eye-Tracking	Dwell Time	120.0 s	< 80.0 s	Extended Fixation
Eye-Tracking	Progress/Regress Ratio	3.8	> 7.0	Reduced Fluency
Speech	Hesitation Rate (>500ms)	4.8 /min	< 2.0 /min	Frequent Pauses
Speech	Oral Reading WPM	64.0 WPM	85 - 110 WPM	Below Grade Level
Cognitive	Phonological Score	52 / 100	> 80 / 100	Support Recommended
Cognitive	Working Memory	58 / 100	> 80 / 100	Mild Delay

2. Explainable AI (XAI) Transparent Attribution

Using SHAP (Shapley Additive exPlanations), the relative impact of each observed factor on the screening outcome is transparently quantified below:

Feature Factor	SHAP Impact	Clinical/Educational Explanation
Speech Hesitation Rate	+0.082	Long inter-word silent gaps indicate difficulty retrieving phonological word representations.
Eye-Gaze Regressions	+0.065	Involuntary saccadic jumps backward in the line indicate decoding breakdown.
Low Phonological Score	+0.058	Challenges in blending individual phonemes into whole words.
Extended Dwell Duration	+0.034	High total fixation time across syllables reflects cognitive reading fatigue.

3. Recommended Personalized Learning Plan

Recommended Daily Practice: 25 minutes | **Primary Goal:** Strengthen phonics decoding and reduce visual reading regressions.

Activity Title	Target Domain	Recommended Daily Time	Priority
Phonics Sound-Blending Quest	Phonological Awareness	12 minutes	Critical
Visual Tracking & Anti-Regression Maze	Oculomotor Reading Flow	8 minutes	High
Rapid Sight-Word Recognition	Reading Fluency	10 minutes	High

Reviewer Guidelines: *This document should be reviewed in consultation with an educational specialist, reading coach, or speech-language pathologist. Continue daily progress tracking within the ReadWise AI application.*

ReadWise AI Research Project — Karunya Institute of Technology & Sciences